

팍리스 실무형 일경험 프로그램

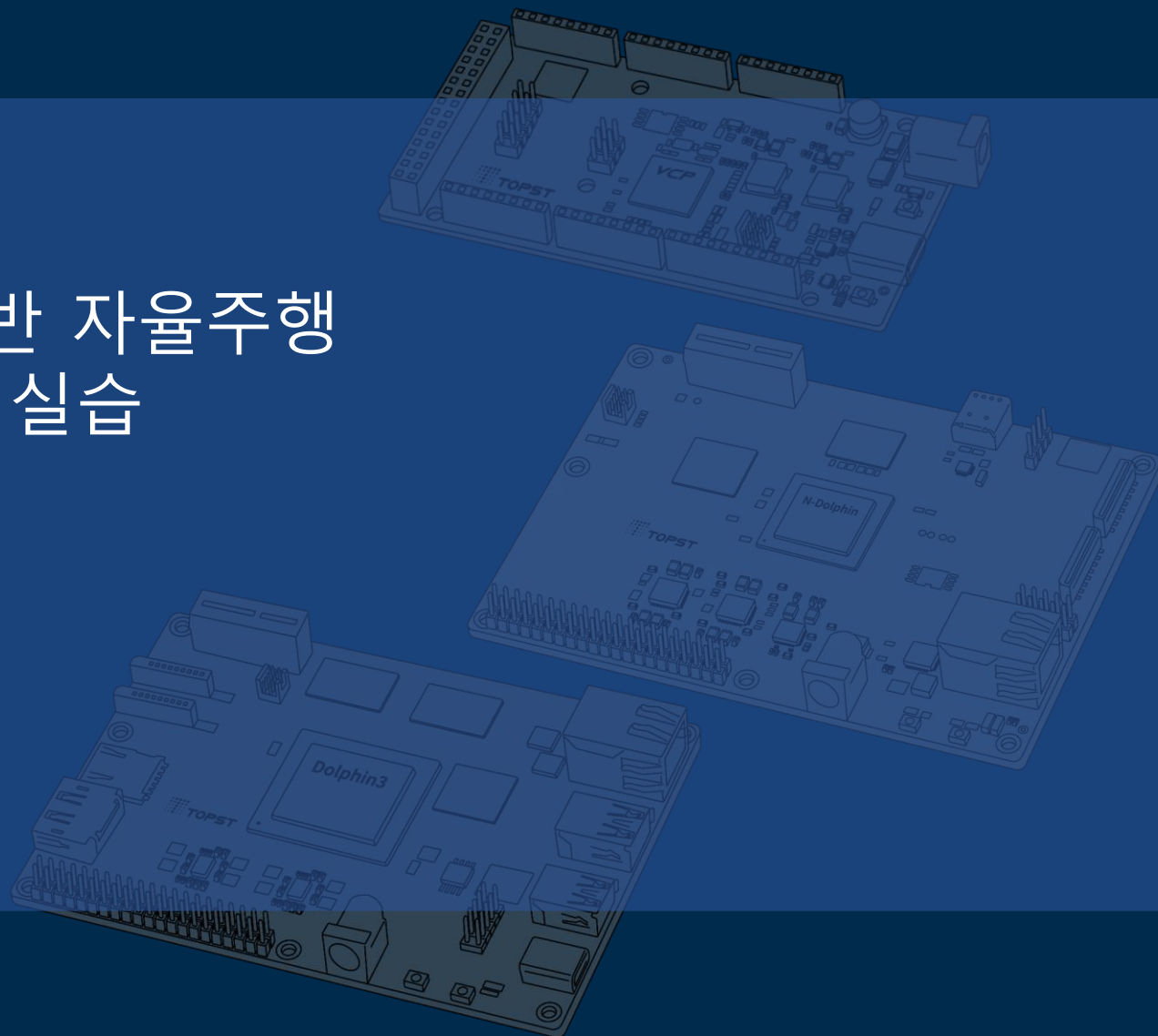
SDV 아키텍처 기반 자율주행 통합 시스템 구현 실습



1일차

05 D3-G 환경 구성

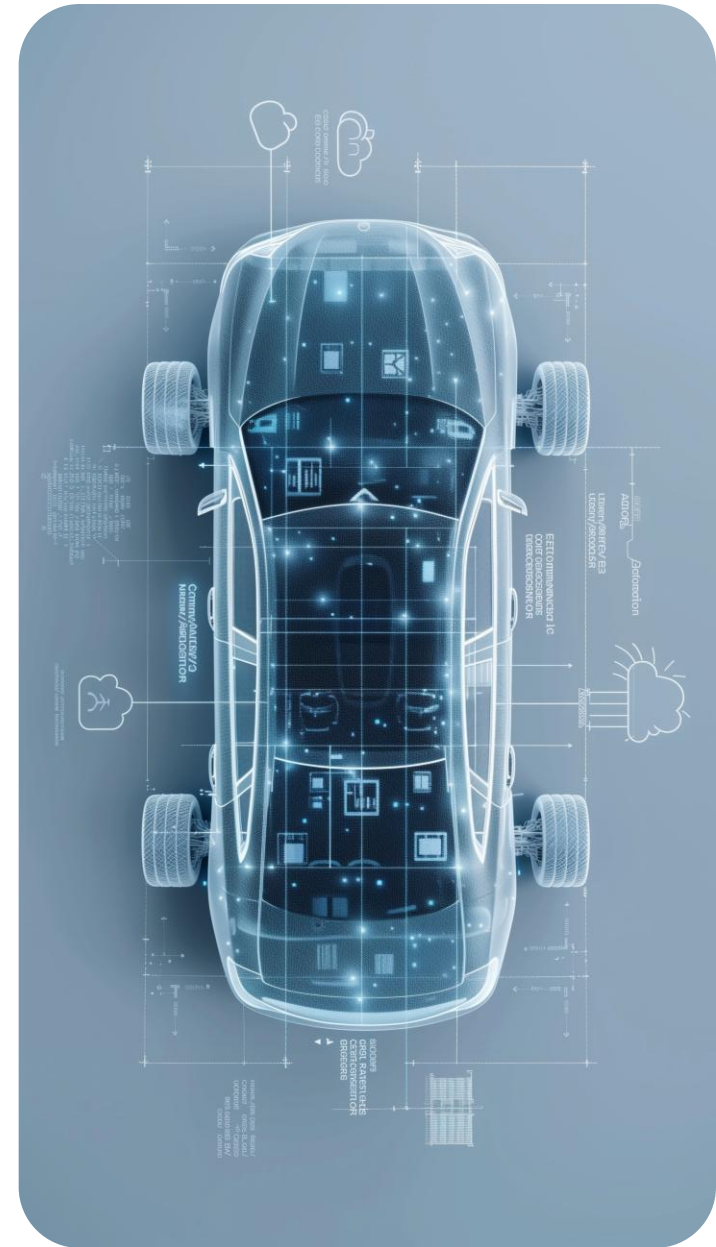
Telechips TOPST



목차

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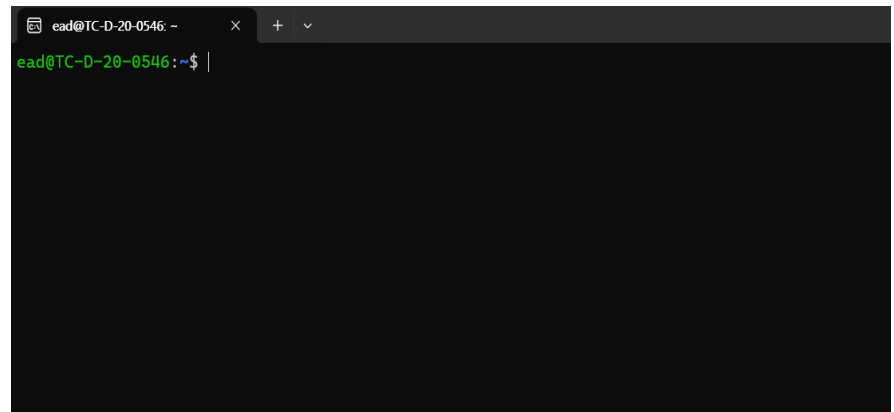
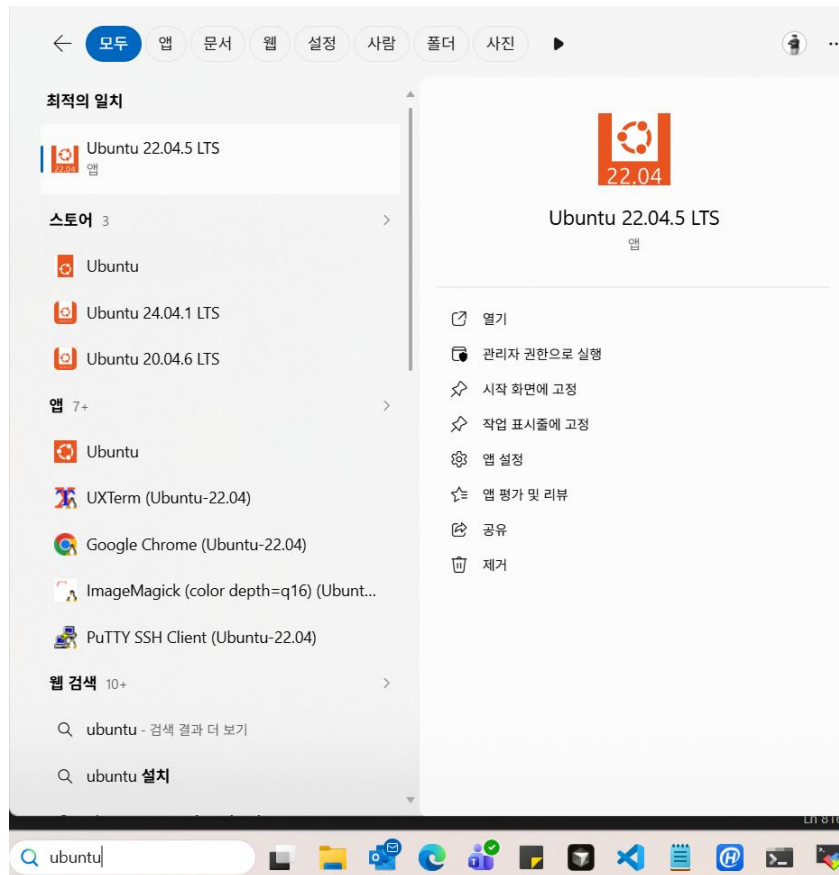
D3-G Specification

Part Name	D3-G
CPU	Arm® Cortex-A72 1.6 GHz, Quad
CPU	Arm® Cortex-A53 1.45 GHz, Quad
CPU	Arm® Cortex-R5
GPU	168 GFLOPS, IMG PowerVR 9XTP
Memory	4 GB (2 GB 32-bit LPDDR4 x 2)
Storage	microSD Card (Not included), 8 GB eMMC 5.1 Flash Storage
Automotive Network	CAN 3-channel
Network	Gigabit Ethernet
Camera	MIPI CSI-2 2-lane, MIPI CSI-2 4-lane
Display	DisplayPort 1.4
USB	USB 2.0 Host, USB 2.0 DRD, USB 3.0 DRD
Others	PCIe Gen3, UART/JTAG debugger port, I2S, I2C, SPI, PWM
Power	5V/3A
OS	Linux
Mechanical	120 mm x 90 mm

- D3-G 는 3개의 코어 존재
 - Cortex-A72
 - Cortex-A53
 - Cortex-R5
- 본 실습에선 **A72 코어와 R5 코어만**을 사용
 - A72 – Ubuntu Weston Desktop 설치
 - R5 – FreeRTOS 설치

D3-G Yocto Build

- WSL Ubuntu 22.04 실행



D3-G Yocto Build

```
$ mkdir -p Work/d3-g-yocto  
$ cd Work/d3-g-yocto  
$ repo init -u https://github.com/topst-development/manifests.git -b release/1.1.0 -m  
linux\_yp4.0\_topst.xml
```

```
kypark@TC-E-20-0417:~/Work/d3-g-yocto$ repo init -u https://github.com/topst-development/manifests.git -b release/1.1.0  
-m linux_yp4.0_topst.xml  
Downloading Repo source from https://gerrit.googlesource.com/git-repo  
remote: Finding sources: 100% (4/4)  
remote: Total 4 (delta 3), reused 4 (delta 3)  
  
... A new version of repo (2.54) is available.  
... New version is available at: /home/kypark/Work/d3-g-yocto/.repo/repo/repo  
... The launcher is run from: /usr/bin/repo  
!!! The launcher is not writable. Please talk to your sysadmin or distro  
!!! to get an update installed.  
  
Your identity is: kiyong.park <kiyong.park@telechips.com>  
If you want to change this, please re-run 'repo init' with --config-name  
  
repo has been initialized in /home/kypark/Work/d3-g-yocto  
kypark@TC-E-20-0417:~/Work/d3-g-yocto$ |
```

D3-G Yocto Build

```
$ repo sync
```

```
$ ls
```

```
kypark@TC-E-20-0417:~/Work/d3-g-yocto$ repo sync
```

```
... A new version of repo (2.54) is available.  
... New version is available at: /home/kypark/Work/d3-g-yocto/.repo/repo/repo  
... The launcher is run from: /usr/bin/repo  
!!! The launcher is not writable. Please talk to your sysadmin or distro  
!!! to get an update installed.
```

```
Syncing: 25% (3/12), done in 32.045s
```

```
Syncing: 100% (12/12) 0:48 | 1 job | 0:14 meta-openembedded @ poky/meta-openembeddedrepo sync has finished successfully.
```

```
kypark@TC-E-20-0417:~/Work/d3-g-yocto$ ls
```

```
easy-setup.sh mktcimg poky stitch-fai-ai.sh stitch-fai-d3.sh tools
```

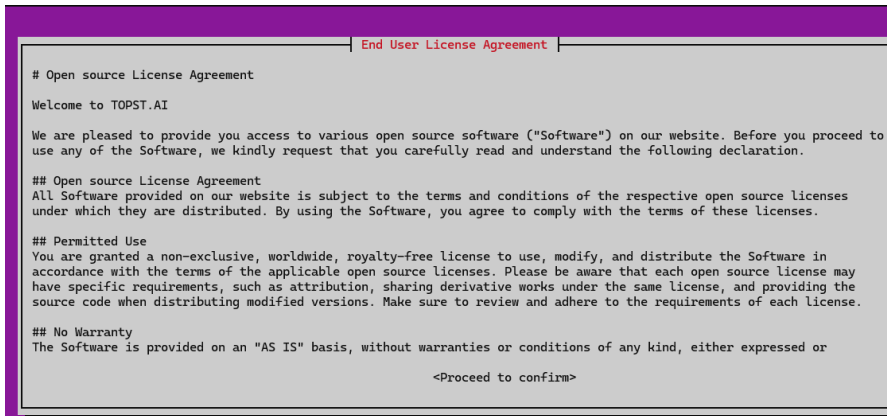
```
kypark@TC-E-20-0417:~/Work/d3-g-yocto$ |
```

D3-G Yocto Build

topst-build.sh 실행

```
$ ./easy-setup.sh
```

스크롤 내리기 -> 방향키(->) 클릭 -> 엔터 -> accept 엔터



```
End User License Agreement

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## No Warranty
The Software is provided on an "AS IS" basis, without warranties or conditions of any kind, either expressed or

<Proceed to confirm>
```



D3-G Yocto Build

topst-build.sh 실행

\$ source poky/meta-topst/topst-build.sh -> 2번 머신 선택 (2 입력)

```
kypark@TC-E-20-0417:~/Work/d3-g-yocto$ source poky/meta-topst/topst-build.sh
Choose MACHINE
  1. ai-g-topst
  2. d3-g-topst-main
  3. d3-g-topst-sub
  4. d5-g-topst-main
  5. d5-g-topst-sub
select number(1-5) => 2
machine(d3-g-topst-main) selected.
You had no conf/local.conf file. This configuration file has therefore been
created for you from /home/kypark/Work/d3-g-yocto/poky/meta-topst/template/d3-g-topst-main/local.conf.sample
You may wish to edit it to, for example, select a different MACHINE (target
hardware). See conf/local.conf for more information as common configuration
options are commented.

You had no conf/bblayers.conf file. This configuration file has therefore been
created for you from /home/kypark/Work/d3-g-yocto/poky/meta-topst/template/d3-g-topst-main/bblayers.conf.sample
To add additional metadata layers into your configuration please add entries
to conf/bblayers.conf.

The Yocto Project has extensive documentation about OE including a reference
manual which can be found at:
  https://docs.yoctoproject.org

For more information about OpenEmbedded see the website:
  https://www.openembedded.org/

Yocto Project common targets are:
  core-image-minimal
  core-image-sato
```


D3-G Yocto Build

Yocto 빌드 수행

```
$ bitbake telechips-topst-image
```

```
kypark@TC-E-20-0417:~/Work/d3-g-yocto/build/d3-g-topst-main$ bitbake telechips-topst-image
NOTE: Your conf/bblayers.conf has been automatically updated.
WARNING: You are running bitbake under WSLv2, this works properly but you should optimize your VHDX file eventually to a
void running out of storage space
Loading cache: 100% | ETA: --:--:--
Loaded 0 entries from dependency cache.
Parsing recipes: 100% |#####| Time: 0:00:24
Parsing of 2382 .bb files complete (0 cached, 2382 parsed). 3770 targets, 325 skipped, 0 masked, 0 errors.
WARNING: No recipes in default available for:
/home/kypark/Work/d3-g-yocto/poky/meta-topst/recipes-connectivity/nanomsg/nng_1.%.bbappend
NOTE: Resolving any missing task queue dependencies

Build Configuration:
BB_VERSION           = "2.0.0"
BUILD_SYS            = "x86_64-linux"
NATIVESBSTRING       = "ubuntu-22.04"
TARGET_SYS           = "aarch64-telechips-linux"
MACHINE              = "d3-g-topst-main"
DISTRO               = "poky-telechips-systemd"
DISTRO_VERSION        = "4.0.17"
TUNE_FEATURES        = "aarch64 armv8a crc cortexa72"
TARGET_FPU           = ""
LINUX_VERSION        = "5.10.205"
KBUILD_DEFCONFIG      = "d3_g_topst_defconfig"
TCC_BSP_FEATURES     = "audio display graphics camera drm network multimedia_vpu_v3"
IMAGE_FEATURES       = " u-boot-net debug-tweaks ssh-server-openssh package-management"
GCCVERSION            = "arm-11.2"
GLIBCVERSION         = "2.35"
TOPST_FEATURES       = " multimedia support-4k-video network support-tty-console support-bt-usb support-nf-docker suppor
t-swap"
```

D3-G Yocto Build

Yocto 빌드 결과 dir 설명

- Kernel : d3-g-topst-main-tc-boot-5.10.205-r0.img, Image--5.10.205-r0-d3-g-topst-main-20250929060303.bin
- Kernel Modules : modules--5.10.205-r0-d3-g-topst-main-20250929060303.tgz
- Device tree bin : tcc8050-topst-d3-g--5.10.205-r0-d3-g-topst-main-20250929060303.dtb
- Rootfs : telechips-topst-image-d3-g-topst-main-20250929060303.rootfs.ext4

```
kypark@TOPST-Build-SVR:~/Work/d3g-github-recent/build/d3-g-topst-main/tmp/deploy/images/d3-g-topst-main$ pwd
/home/kypark/Work/d3g-github-recent/build/d3-g-topst-main/tmp/deploy/images/d3-g-topst-main
kypark@TOPST-Build-SVR:~/Work/d3g-github-recent/build/d3-g-topst-main/tmp/deploy/images/d3-g-topst-main$ ls
ca72_bl3-d3-g-topst-main-1.0-r0.rom          modules-d3-g-topst-main.tgz
ca72_bl3-d3-g-topst-main.rom               tc-boot-d3-g-topst-main.img
ca72_bl3.rom                               tc-boot.img
d3-g-topst-main-tc-boot-5.10.205-r0.img     tcc8050-topst-d3-g--5.10.205-r0-d3-g-topst-main-20250929060303.dtb
Image                                       tcc8050-topst-d3-g-d3-g-topst-main.dtb
Image--5.10.205-r0-d3-g-topst-main-20250929060303.bin
Image-d3-g-topst-main.bin                  tcc8050-topst-d3-g.dtb
Image-initramfs--5.10.205-r0-d3-g-topst-main-20250929060303.bin
Image-initramfs-d3-g-topst-main.bin         telechips-topst-image-d3-g-topst-main-20250929060303.rootfs.ext4
initramfs-topst-image-d3-g-topst-main-20250929060303.rootfs.cpio
initramfs-topst-image-d3-g-topst-main-20250929060303.rootfs.manifest
initramfs-topst-image-d3-g-topst-main-20250929060303.testdata.json
initramfs-topst-image-d3-g-topst-main.cpio  telechips-topst-image-d3-g-topst-main-20250929060303.rootfs.manifest
initramfs-topst-image-d3-g-topst-main.manifest
initramfs-topst-image-d3-g-topst-main.testdata.json
modules--5.10.205-r0-d3-g-topst-main-20250929060303.tgz
kypark@TOPST-Build-SVR:~/Work/d3g-github-recent/build/d3-g-topst-main/tmp/deploy/images/d3-g-topst-main$
```

D3-G Yocto 이미지

fwdn 이미지 추출

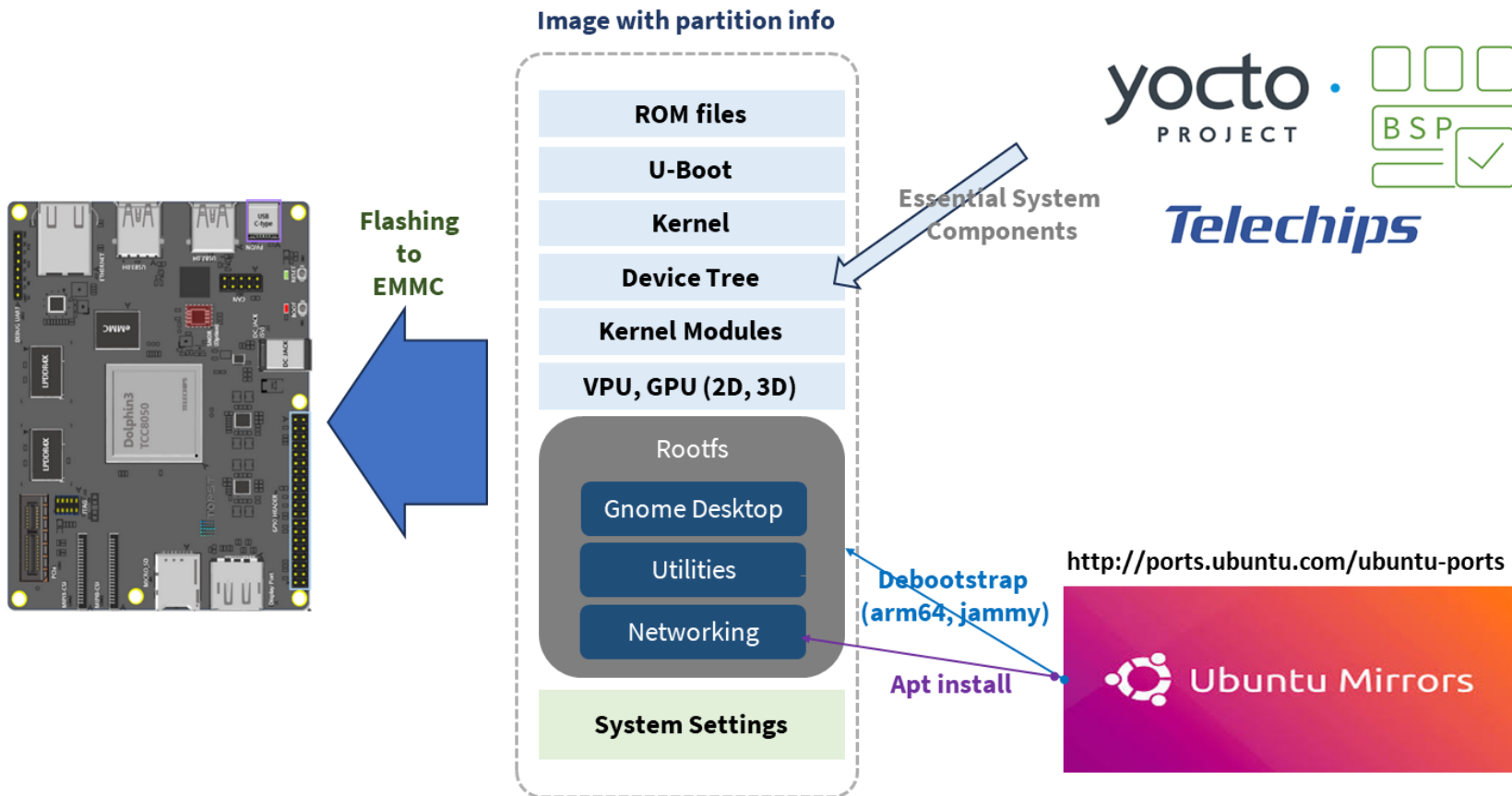
```
$ cd ~/Work/d3-g-yocto
```

```
$ ./stitch-fai-d3.sh -f => output_d3g.fwdn.zip 파일 생성 (fw 설치 위한 필요 파일 압축본)
```

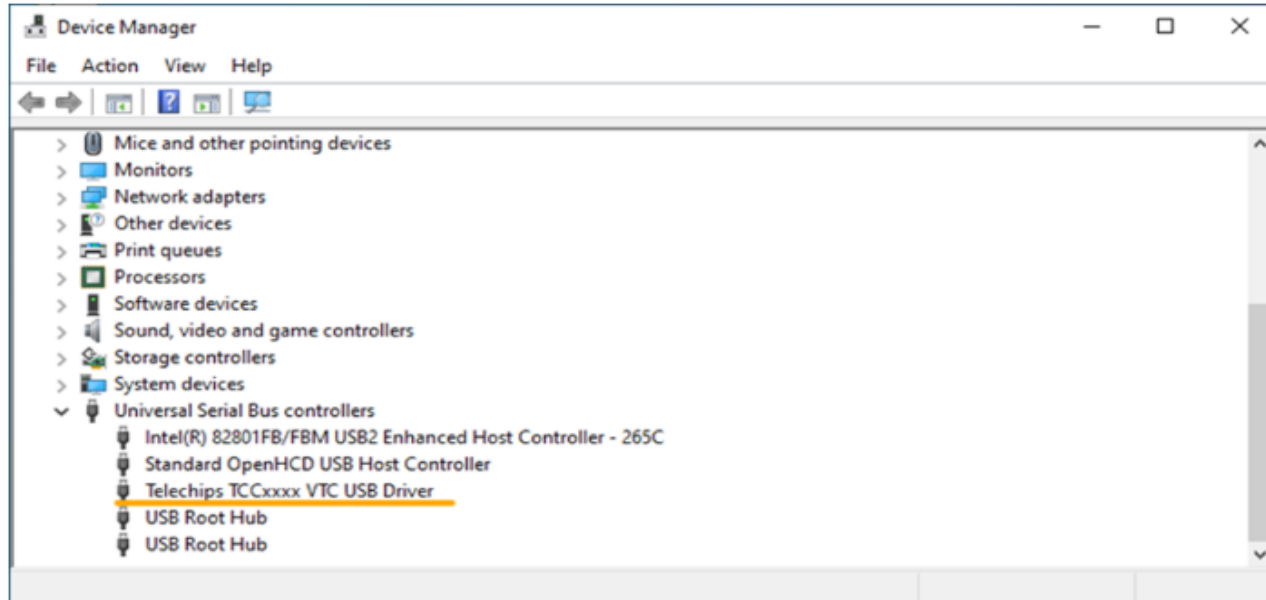
```
kypark@T0PST-Build-SVR:~/Work/d3g-github-recent$ ls -al
total 216380
drwxrwxr-x  7 kypark kypark      4096 Nov  3 09:28 .
drwxrwxr-x 60 kypark kypark      4096 Oct 16 13:51 ..
drwxrwxr-x  3 kypark kypark      4096 Sep 29 15:02 build
lrwxrwxrwx  1 kypark kypark        19 Sep 29 15:01 easy-setup.sh -> tools/easy-setup.sh
-rw-rw-r--  1 kypark kypark         0 Sep 29 15:02 .eula_accepted
drwxrwxr-x  4 kypark kypark      4096 Sep 29 15:01 mktcimg
-rw-rw-r--  1 kypark kypark 221536745 Nov  3 09:28 output_d3g.fwdn.zip
drwxrwxr-x 20 kypark kypark      4096 Sep 29 15:01 poky
drwxrwxr-x  7 kypark kypark      4096 Sep 29 15:01 .repo
lrwxrwxrwx  1 kypark kypark        22 Sep 29 15:01 stitch-fai-ai.sh -> tools/stitch-fai-ai.sh
lrwxrwxrwx  1 kypark kypark        22 Sep 29 15:01 stitch-fai-d3.sh -> tools/stitch-fai-d3.sh
drwxrwxr-x  4 kypark kypark      4096 Sep 29 15:01 tools
kypark@T0PST-Build-SVR:~/Work/d3g-github-recent$
```

D3-G Ubuntu 이미지

Ubuntu Image 다운로드 : [D3-G Gnome image](#)



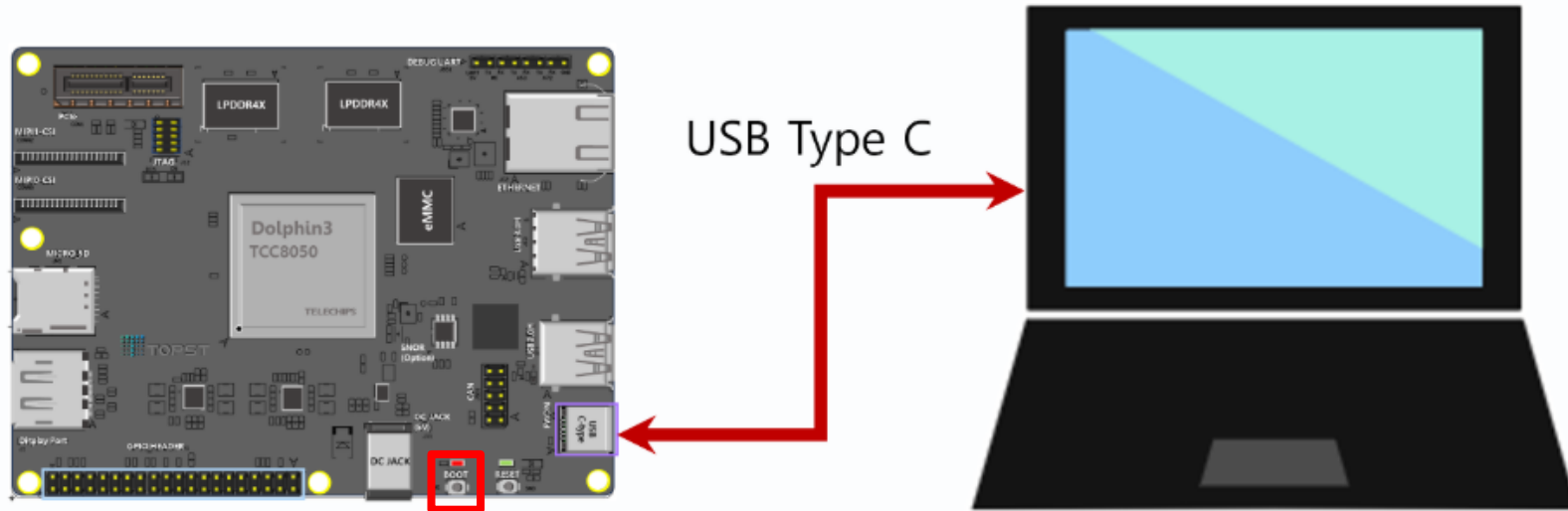
D3-G 환경 설정(공통)



- **Install VTC Driver**

- 펌웨어 다운로드를 위해 VTC(Vendor Telechips Certification) 드라이버 설치 필수
- 관리자 권한으로 실행: [VTC 드라이버](#)
- 드라이버 설치 후 FWDN 모드에서 USB 연결 후 설정 확인

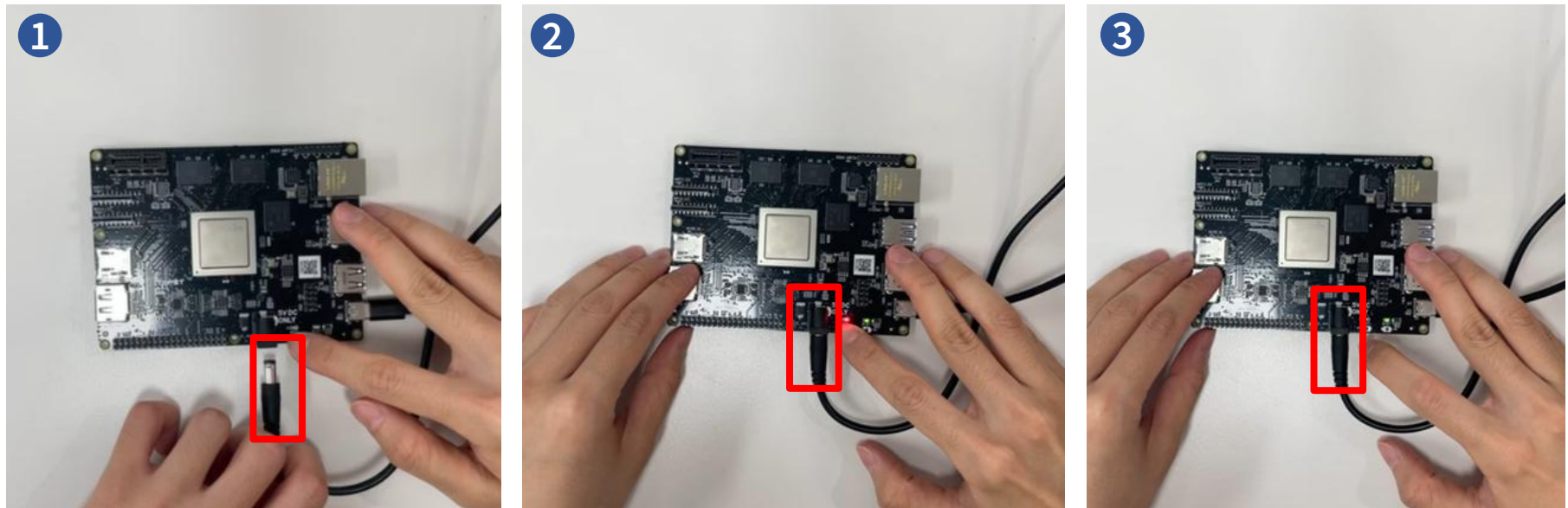
D3-G 환경 설정 (공통)



- **Connect D3-G Board and Host PC with USB Boot Mode**

- USB Boot Mode로 들어가기 위해 FWDN 스위치를 누른 상태에서 전원 케이블을 D3-G 보드에 연결
- USB Type-C 케이블을 D3-G 보드 및 Host PC의 USB Type-C FWDN 포트에 연결
- D3-G : **C 케이블** & Host PC : **USB 케이블**

D3-G 환경 설정 (공통)



- **Boot Mode 진입**

- BOOT 스위치를 누른 채로, 전원 플러그(5V) 연결(BOOT Switch LED 상태 : ON)
- 이후 스위치에서 손가락을 떼면 진입 완료(BOOT Switch LED 상태 : OFF)

D3-G A72 이미지 업로드 (공통)

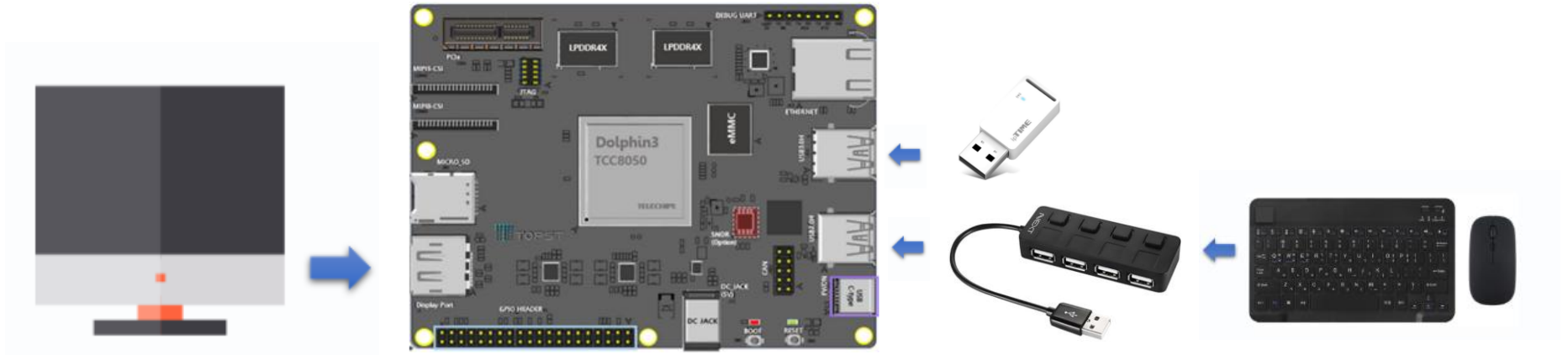


• FWDN (D3-G Ubuntu Weston/Gnome Desktop)

- 제공된 d3g-ubuntu zip 파일을 PC Windows 혹은 PC ubuntu에 다운로드 후 압축 해제
- Ex) Windows
 - 받은 폴더 내에서 마우스 우클릭> 터미널에서 열기(명령프롬프트) 실행
 - fwdn.bat (Windows 배치 파일)실행
 - fwdn.bat은 펌웨어를 자동으로 다운로드 하는 실행 파일
 - \$.\fwdn.bat

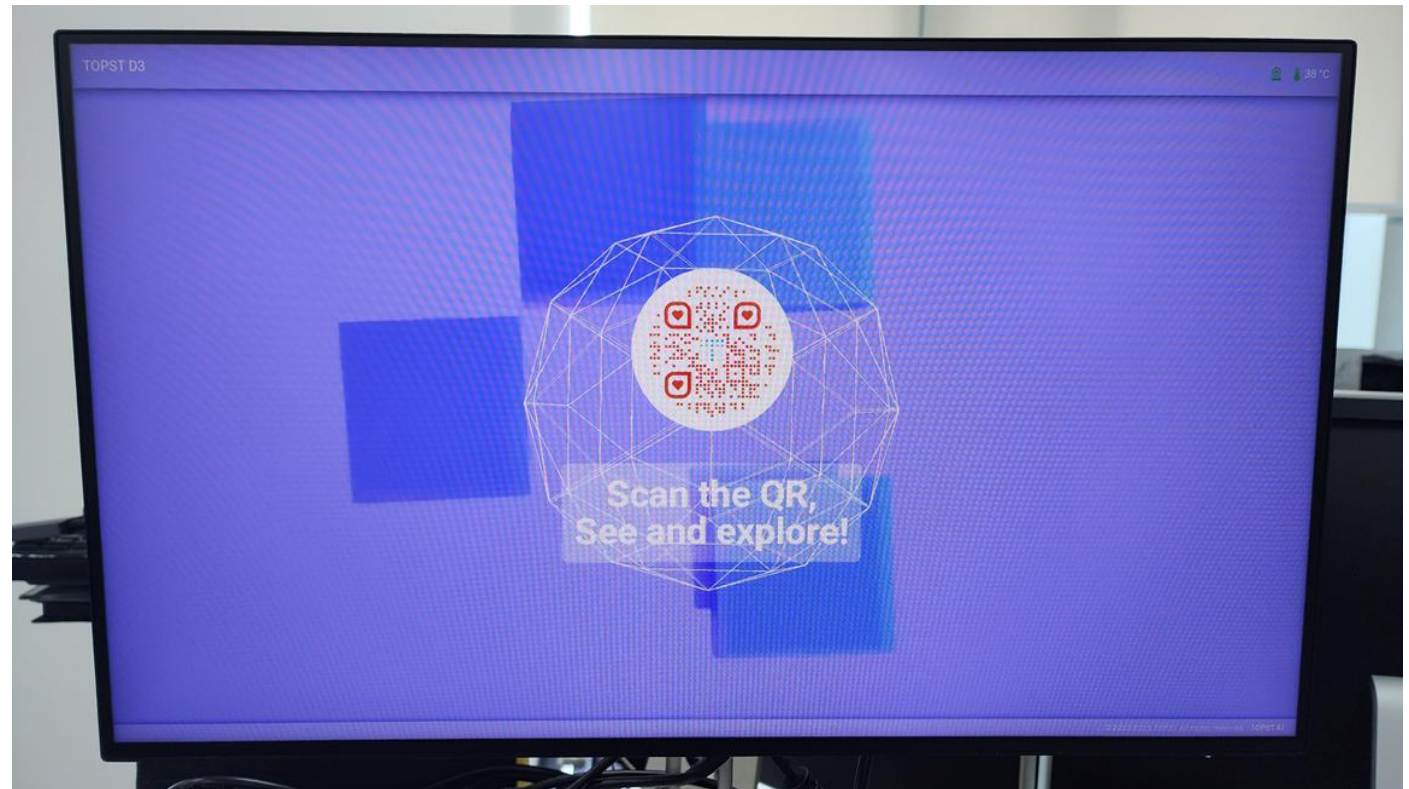
D3-G A72 이미지 (Yocto Weston)

- 설치되는 동안 보드에 모니터와 키보드, 마우스, wifi 모듈 연결



D3-G A72 이미지 (Yocto Weston)

- 설치가 완료되면 전원 재인가 or reset 버튼 누르기
- 암호 “topst” 입력



Demo Web-site on localhost

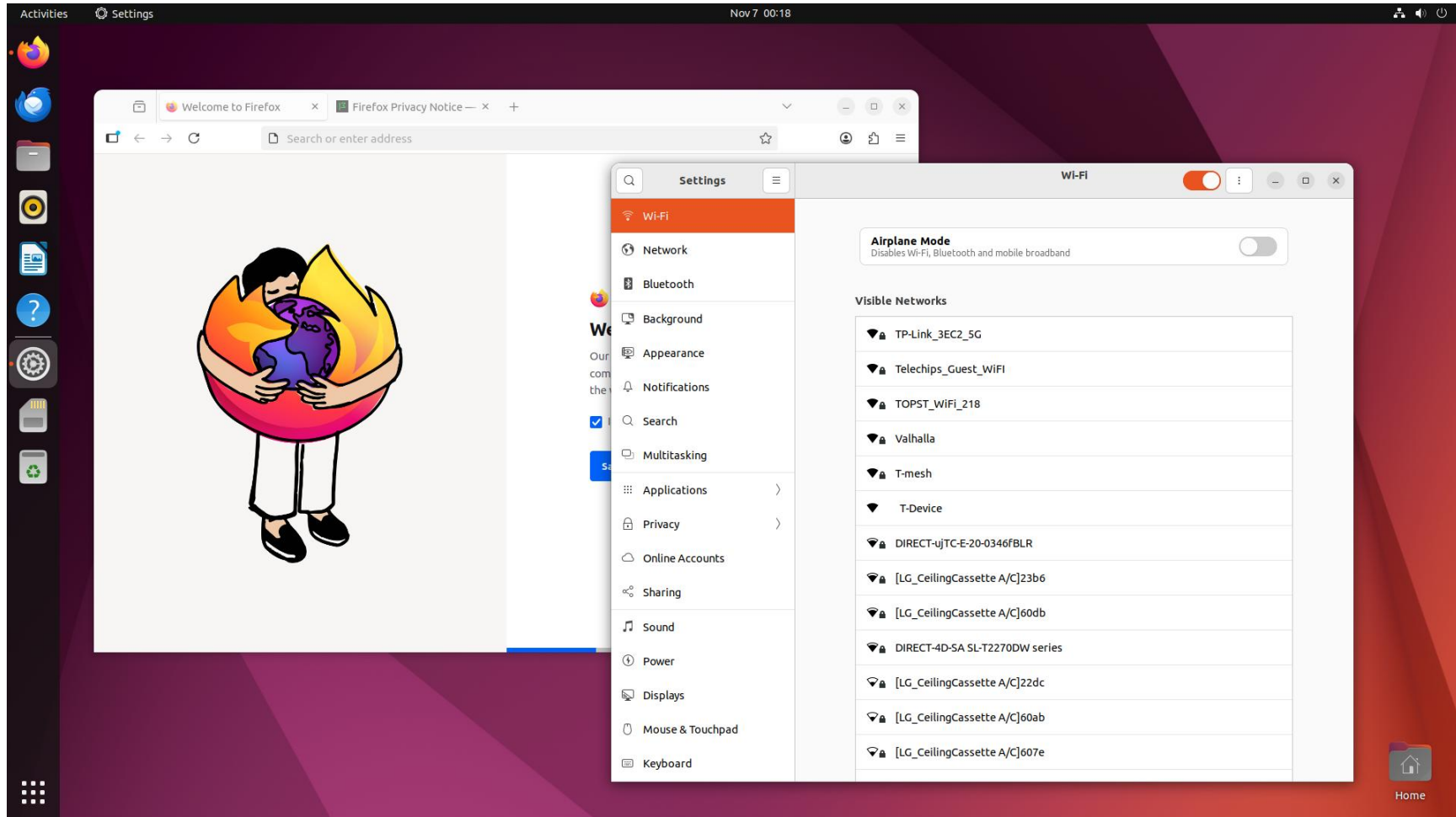
COG browser

WPE Webkit

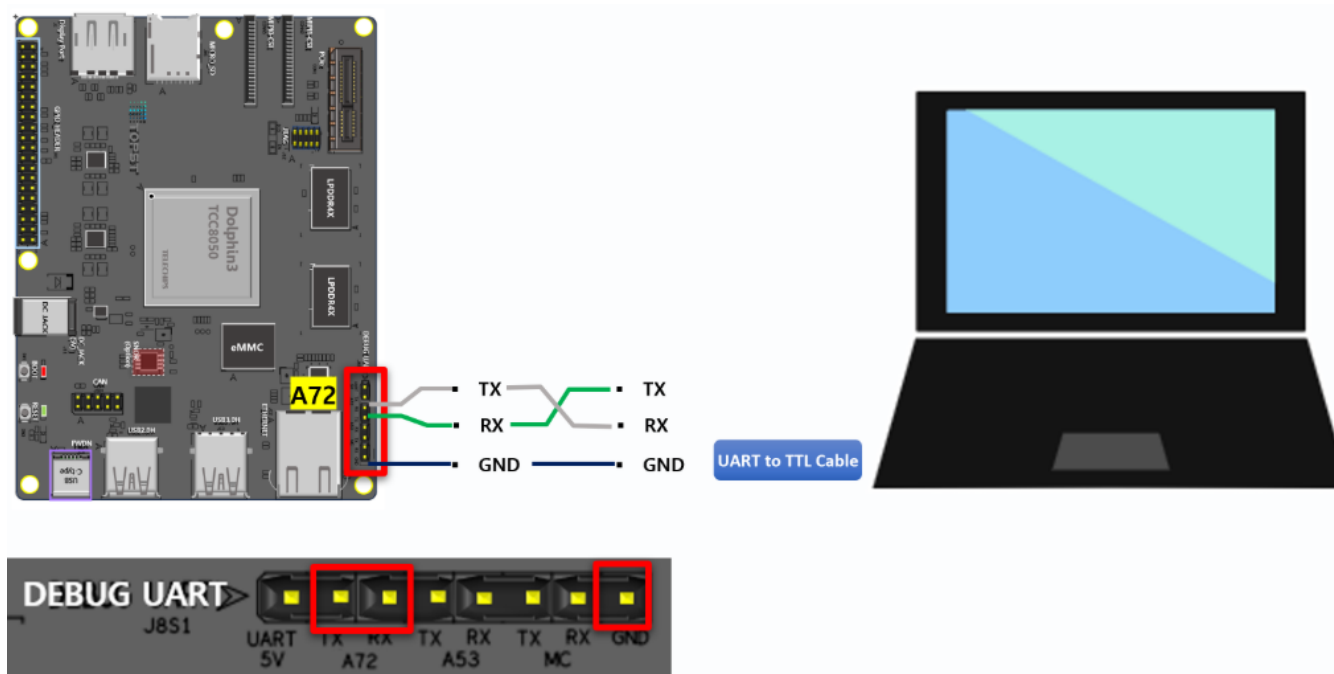
WPE Backend

D3-G system

D3-G A72 이미지 (Ubuntu Gnome)

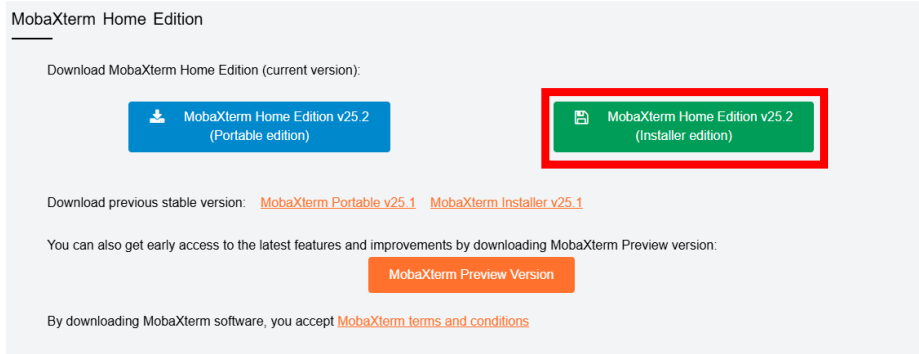


D3-G A72 ↔ PC 통신



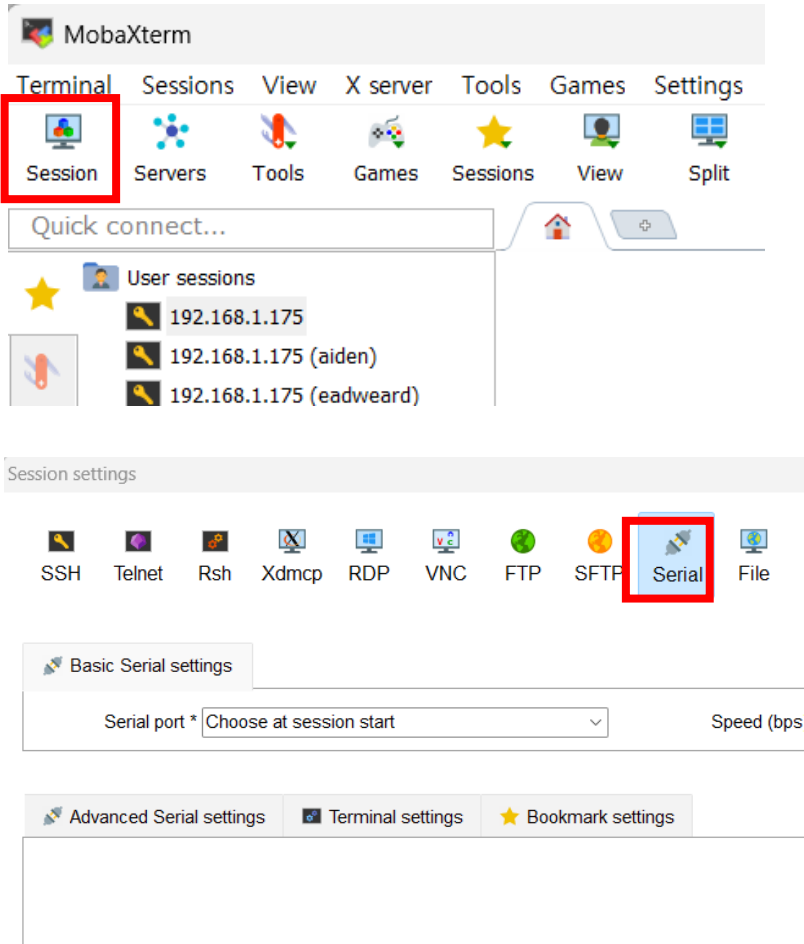
- **Connect D3-G A72 and Host PC with UART**
 - D3-G A72 코어와 PC 간 USB-to-TTL cable을 이용하여 연결
 - 검정 케이블은 UART 핀의 **GND**에 연결
 - 흰 케이블은 UART 핀의 **TX** 핀에 연결 & 녹색 케이블은 UART 핀의 **RX** 핀에 연결

D3-G A72 ↔ PC 통신

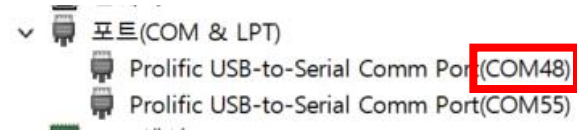


- **Connect D3-G Board and Host PC with UART**
 - Serial port driver 설치
 - [CP210x Universal Windows Driver](#)
 - [PL2303_prolific Driver](#)
 - **Terminal emulator 설치 (MobaXterm)**
 - [MobaXterm Xserver with SSH, telnet, RDP, VNC and X11 - Download](#) - 링크 접속
 - **압축 해제**
 - 설치한 MobaXterm_Installer_v25.2 압축 해제
 - MobaXterm_installer_25.2 실행파일 실행
-  MobaXterm_installer.dat
-  MobaXterm_installer_25.2
- Next 두는 우 설치 실행

D3-G A72 ↔ PC 통신



- 윈도우 검색창에 MobaXterm 검색 후 실행
- 오른쪽 상단에 Session 버튼 클릭
- Serial 클릭
- 장치 관리자를 통해 port 확인



D3-G A72 ↔ PC 통신

The screenshot shows the MobaXterm configuration window. The 'Basic Serial settings' tab is active, with 'Serial port' set to 'COM4 (Prolific USB-to-Serial Comm Port(CO...' and 'Speed (bps)' set to '115200'. The 'Advanced Serial settings' tab is also visible, showing 'Serial engine' as 'PuTTY (allows manual COM port setting)', 'Data bits' as '8', 'Stop bits' as '1', 'Parity' as 'None', and 'Flow control' as 'None'. A red box highlights the 'Flow control' dropdown. Below these settings are buttons for 'Reset defaults' and a dropdown for 'Execute macro at session start' set to '<none>'. A text box on the right provides instructions on using MobaXterm's embedded TFTP server.

Basic Serial settings

Serial port * COM4 (Prolific USB-to-Serial Comm Port(CO... Speed (bps) * 115200

Advanced Serial settings Terminal settings Bookmark settings

Serial engine: PuTTY (allows manual COM port setting)

Data bits 8

Stop bits 1

Parity None

Flow control None

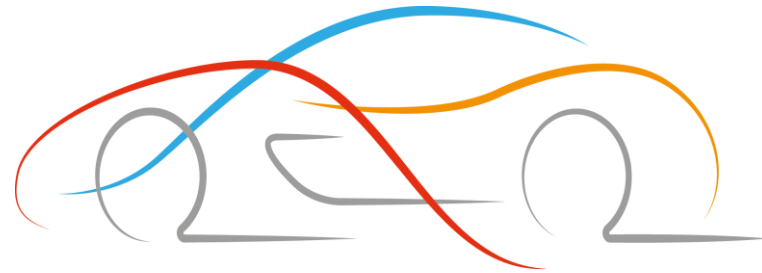
Reset defaults

Execute macro at session start: <none>

If you need to transfer files (e.g. router configuration file), you can use MobaXterm embedded TFTP server

"Servers" window --> TFTP server

- Serial port에서 연결된 포트 설정
 - Speed 115200 설정
- Advanced Serial settings 탭 선택
 - Flow control에서 None 선택
- ok눌러 보드 접속
- ID : topst / PW : topst 입력



Thank You